

ASSIGNMENT-14.3

HT.NO-2203A52100

BATCH-50

QUESTION:

Lab 14: Web Design Applications Using AI Assistance

HTML, CSS, and JavaScript Generation with GitHub Copilot

Lab Objectives

The objectives of this laboratory exercise are to:

Week7 - Wednesday

- Design functional and visually appealing web applications using HTML, CSS, and JavaScript.
- Use AI-assisted tools to accelerate front-end development.
- Apply responsive, accessible, and interactive web design principles.
- Customize and validate AI-generated UI code for real-world applications.

Learning Outcomes

After completing this lab, students will be able to:

- Generate structured HTML layouts using AI assistance.
- Apply responsive CSS techniques such as Flexbox and Grid.
- Implement interactive client-side features using JavaScript.
- Critically review and improve AI-generated front-end code.

Tools Required

- Visual Studio Code (VS Code)
- GitHub Copilot Extension
- Web Browser (Chrome / Firefox)
- HTML5, CSS3, JavaScript (ES6)

Task 1: AI-Assisted Personal Portfolio Website

Scenario

A student wants to showcase academic projects, technical skills, and contact information through a personal portfolio website. AI tools are used to speed up development while maintaining quality and customization.

Tasks

1. Use GitHub Copilot to generate a semantic HTML structure with sections:

o Home o

About o

Projects o

Contact

2. Ask Copilot to suggest responsive CSS using Flexbox or Grid.

3. Customize AI-generated CSS to add:

o Hover effects on project cards o

Smooth scrolling navigation

Expected Outcome

- Clean and well-structured portfolio website
- Responsive layout for desktop and mobile screens • Interactive project cards

Task 2: AI-Generated Restaurant Landing Page

Scenario

A local restaurant requires a simple yet attractive landing page to display its offerings. The developer uses AI assistance to rapidly generate UI components.

Tasks

1. Use Copilot to create a navigation bar with links:

o Home

o Menu o

Gallery o

Contact

2. Generate a menu section styled with CSS cards.

3. Implement a JavaScript-based image slider for the gallery using Copilot suggestions.
4. Customize animations or transitions for better user experience.

Expected Outcome

- Fully functional restaurant landing page
- Responsive navigation bar
- Working image slider with smooth transitions

Task 3: AI-Powered Event Registration Form

Scenario

SR University is hosting a technical fest and requires an online registration form with real-time validation to ensure accurate user input.

Tasks

1. Ask Copilot to generate an HTML form with the following fields:

o Name o Email

o Phone Number

o Department o

Event Selection

2. Use AI assistance to style the form using CSS for readability and accessibility.
3. Implement JavaScript validation using Copilot: o Email format validation o Phone number length validation o Required field checks

Expected Outcome

- User-friendly and visually appealing form
- Real-time validation messages
- Prevention of invalid submissions

Task 4: AI-Assisted E-Commerce Product Page

Scenario

A startup is developing a basic e-commerce product page to showcase products and manage simple cart functionality.

Tasks

1. Use Copilot to generate a grid-based product catalog using HTML and CSS.
2. Implement an “Add to Cart” feature using JavaScript.
3. Modify AI-generated code to include:
 - o Cart item counter at the top-right corner
 - o Visual feedback when a product is added

Expected Outcome

- Functional product catalog
- Working cart counter
- Interactive user experience

Task 5: Create a Responsive Web Page Layout

Instructions:

- Design a basic web page layout with a header, main content area, and footer using HTML and CSS.
- Use AI to assist in generating responsive CSS for different screen sizes.
- Ensure the layout is clean and visually organized.

Expected Output:

- A responsive web page with:
- Header with navigation links
- Main section with placeholder text/images
- Footer with copyright or contact info
- Layout adapts correctly to desktop, tablet, and mobile screen sizes.

CODE:

```
import os
```

```
import webbrowser
```



```
# Create folder folder =
```

```
"Lab14_Web_AI"
```

```
os.makedirs(folder, exist_ok=True)
```

```
# ----- TASK 1: PORTFOLIO ----- portfolio_html
```

```
= """<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Portfolio</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<nav>
```

```
<a href="#home">Home</a>
```

```
<a href="#about">About</a>
```

```
<a href="#projects">Projects</a>
```

```
<a href="#contact">Contact</a>
```

```
</nav>
```

```
<section id="home"><h1>Welcome to My Portfolio</h1></section>
```

```
<section id="about"><h2>About Me</h2><p>Student developer.</p></section>
```

```
<section id="projects">
```

```
<h2>Projects</h2>
```

```
<div class="cards">
```

```
<div class="card">Project 1</div> <div
```

```
class="card">Project 2</div>
```

</div>

</section>

<section id="contact">

<h2>Contact</h2>

<p>Email: example@email.com</p>

</section>

</body>

</html>

"""

----- TASK 2: RESTAURANT ----- restaurant_html

= """<!DOCTYPE html>

<html>

<head>

<title>Restaurant</title>

<link rel="stylesheet" href="style.css">

</head>

<body>

<nav>

Home

Menu

Gallery

Contact

</nav>

<h1>Restaurant</h1>


```
<div class="cards">
```

```
<div class="card">Pizza</div>
```

```
<div class="card">Burger</div>
```

```
</div>
```

```
<div class="slider">
```

```

```

```
</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

```
"""
```

```
# ----- TASK 3: FORM ----- form_html
```

```
= """<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Event Form</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<h2>Registration Form</h2>
```

```
<form onsubmit="return validateForm()">
```

```
<input type="text" id="name" placeholder="Name" required><br>
```

```
<input type="email" id="email" placeholder="Email" required><br>
<input type="text" id="phone" placeholder="Phone" required><br>
<button type="submit">Submit</button>

</form>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

```
"""
```

```
# ----- TASK 4: E-COMMERCE ----- ecommerce_html
```

```
= """<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>E-Commerce</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<h1>Products</h1>
```

```
<div id="cart">Cart: <span id="count">0</span></div>
```

```
<div class="cards">
```

```
<div class="card">
```

```
<p>Product 1</p>
```

```
<button onclick="addToCart()">Add</button>
```

```
</div>
```

```
<div class="card">
```



```
<p>Product 2</p>
<button onclick="addToCart()">Add</button>

</div>

</div>
```

```
<script src="script.js"></script>
```

```
</body>
```

```
</html>
```

```
"""
```

```
# ----- TASK 5: BASIC LAYOUT ----- layout_html
```

```
= """<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<title>Layout</title>
```

```
<link rel="stylesheet" href="style.css">
```

```
</head>
```

```
<body>
```

```
<header>Header</header>
```

```
<main>Main Content</main>
```

```
<footer>Footer</footer>
```

```
</body>
```

```
</html>
```

```
"""
```

```
# ----- COMMON CSS ----- css = ""
```

```
body {font-family: Arial; margin: 0;} nav {background:  
black; padding: 10px;} nav a {color: white; margin: 10px;  
text-decoration: none;}
```

```
.cards {display: flex; gap: 10px;}
```

```
.card { padding: 20px;
```

```
background: lightgray;
```

```
transition: 0.3s;
```

```
}
```

```
.card:hover {background: gray; color: white;}
```

```
header, footer {
```

```
background: black;
```

```
color: white; text-align:
```

```
center; padding: 10px;
```

```
}
```

```
main {padding: 20px;}
```

```
@media(max-width:600px){
```

```
.cards {flex-direction: column;}
```

```
}
```

```
""
```

```
# ----- COMMON JS -----
```

```
js = "" let count = 0;
```



```
function addToCart(){ count++;
document.getElementById("count").innerText = count;
alert("Added to cart!");
}
```

```
function validateForm(){ let phone =
document.getElementById("phone").value;
if(phone.length != 10){ alert("Phone must be 10
digits"); return false; } return true;
}
```

```
// Slider let
images = [
"https://via.placeholder.com/400",
"https://via.placeholder.com/400/0000FF",
"https://via.placeholder.com/400/FF0000"
];
```

```
let i = 0;
setInterval(()=>{ let slide =
document.getElementById("slide"); if(slide){
i = (i+1)%images.length; slide.src =
images[i];
}
},2000);
""
```

```
# ----- SAVE FILES ----- files
= {
```

```

"portfolio.html": portfolio_html,
"restaurant.html": restaurant_html,
"form.html": form_html,
"ecommerce.html": ecommerce_html,
"layout.html": layout_html,
"style.css": css,
"script.js": js
}

```

```

for name, content in files.items():    with open(os.path.join(folder,
name), "w", encoding="utf-8") as f:
    f.write(content)

```

```

# ----- OPEN ONE PAGE -----

```

```

webbrowser.open(os.path.abspath(os.path.join(folder, "portfolio.html")))

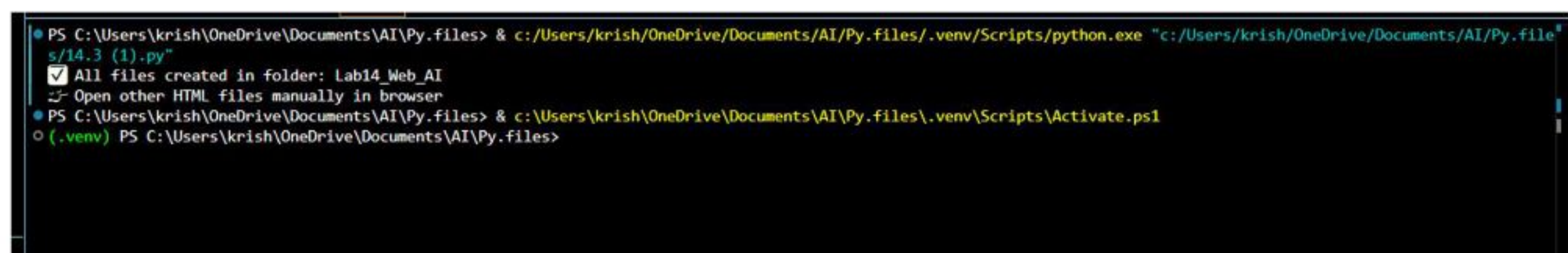
```

```

print(" All files created in folder:", folder) print(" Open
other HTML files manually in browser")

```

OUTPUT:



```

PS C:\Users\krish\OneDrive\Documents\AI\Py.files> & c:/Users/krish/OneDrive/Documents/AI/Py.files/.venv/Scripts/python.exe "c:/Users/krish/OneDrive/Documents/AI/Py.files/s/14.3 (1).py"
[✓] All files created in folder: Lab14_Web_AI
[-] Open other HTML files manually in browser
PS C:\Users\krish\OneDrive\Documents\AI\Py.files> & c:\Users\krish\OneDrive\Documents\AI\Py.files\.venv\Scripts\Activate.ps1
(.venv) PS C:\Users\krish\OneDrive\Documents\AI\Py.files>

```